Harsha Hiredesai

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Education

Georgia Institute of Technology — Expected May 2028
B.S. in Computer Engineering (Threads: Cybersecurity + Distributed Systems & Software Design) — *Atlanta, GA*
- **Relevant Coursework:** Data Structures and Algorithms, Digital System Design
- **Transferred** from Arizona State University, B.S. Computer Science (Cybersecurity) | GPA: 3.8

Experience

AI Security Engineering Intern — May 2026 – Aug 2026
Numberz.ai — *Remote*
- Built and owned development of an 18-payload LLM red-team exploit library covering prompt injection and sensitive information disclosure, run against an authorized dev instance of a governed NL-to-SQL platform; findings mapped to NIST 800-53 and NIST AI RMF.
- Wrote a Python attack connector that replays payloads against the platform's SSE chat API and exports JSON/XLSX results, turning one-off manual tests into repeatable regression runs.
- Reported 2 confirmed findings: leakage of the system prompt and internal architecture, and database schema enumeration that succeeded even after the model refused direct injection attempts.
- Extended coverage to indirect prompt injection via poisoned documents and authorization-bypass sweeps against the platform's RBAC layer; packaged the library as a CLI scanner with JSON and Markdown reports.
- Built a Dockerized Python CLI audit tool combining Syft (SBOM), Bandit (SAST), Gitleaks (secrets), and the NVD API (CVE lookups) into a single vulnerability report.
- Added Zod input validation and a rate-limiting design across 7 API routes in a TypeScript/Express backend with vitest/supertest coverage; authored a STIG hardening checklist reviewed by senior engineers for GovCon compliance.

Cybersecurity / Cloud Engineer Intern — Dec 2025 – Mar 2026
The Commons XR — *Remote*
- Configured GCP VPC networks and VPC peering for cross-network connectivity, documenting the architecture and security configuration for team review.

Software Developer Intern — May 2025 – Aug 2025
GradAchiever — *Remote*
- Developed Python automation scripts using BeautifulSoup for large-scale data scraping and normalization, and ran regression testing on WordPress plugin updates to keep the platform stable.

Projects

Continuous LLM Security Red-Teaming Platform | *Python, FastAPI, LLM Security (OWASP LLM Top 10)*
- Built an open-source LLM red-teaming platform in Python covering 5 attack classes — direct and indirect prompt injection, jailbreaks, system-prompt extraction, and data exfiltration — in a plugin probe/detector architecture.
- Designed a two-signal detection engine that confirms a finding only when an LLM judge and a deterministic check agree, cutting false positives versus keyword matching.
- Built a deliberately vulnerable FastAPI RAG app as a test target, with canary tokens confirming 100% detection of executed injections.
- Wired the harness into GitHub Actions as a security gate that fails builds on confirmed model-behavior regressions, with JSON/HTML reports mapped to the OWASP LLM Top 10.

Secure CI/CD Pipeline | *GitHub Actions, Terraform, AWS, Docker, OPA/Rego, Python*
- Built a shift-left DevSecOps pipeline with 5 automated security gates — Gitleaks (secrets), Bandit (SAST), pip-audit (SCA), OPA/Conftest (IaC policy), and Trivy (containers) — blocking vulnerable builds before deployment.
- Implemented least-privilege AWS posture in Terraform: OIDC role assumption for GitHub Actions (removing long-lived credentials), KMS encryption for ECR/S3, immutable image tags, and CloudTrail audit logging.
- Wrote 8 OPA/Rego policies enforcing S3 access controls, encryption-at-rest, IAM wildcard deny, and KMS key rotation against Terraform plan JSON; added pre-commit hooks and SARIF upload to the GitHub Security tab.

Web Application Vulnerability Scanner | *Python, aiohttp, asyncio, BeautifulSoup*
- Built an async scanner detecting SQLi, XSS, IDOR, and broken authentication through automated crawling and payload injection against DVWA and Juice Shop.
- Cut false positives with DB-fingerprint matching for SQLi, canary reflection checks for XSS, and response diffing for IDOR; reports rank findings by severity and map to OWASP identifiers.

AWS IAM Misconfiguration Scanner | *Python, Boto3, Flask, AWS IAM*
- Built a Python tool auditing AWS IAM for over-permissive roles, inactive users, and missing MFA, with detection logic for 8 privilege escalation paths mapped to MITRE ATT&CK (TA0004) including PassRole abuse.
- Surfaced findings through a Flask/JS interface so remediation can be prioritized by severity.

Certifications

CompTIA Security+ (SY0-701) | Google Cybersecurity Professional Certificate | TestOut Security PRO

Technical Skills

Languages: Python (Boto3, FastAPI, Flask, Pydantic), TypeScript/JavaScript, SQL, Bash, PowerShell, Java, C/C++, HTML, CSS
Security & Compliance: LLM/AI red-teaming, prompt injection, OWASP Top 10 & OWASP LLM Top 10, IAM, MITRE ATT&CK, Zero Trust, threat modeling, CIS Benchmarks, NIST CSF, NIST AI RMF, STIG, SAST/SCA, DevSecOps, SBOM, secrets management, web application security, security protocols, secure network design
Security Tools: Garak, Promptfoo, Gitleaks, Bandit, Trivy, Semgrep, pip-audit, Syft, Checkov, Wireshark, Splunk
Cloud & DevOps: AWS (IAM, EC2, S3, Lambda, ECR, KMS, CloudTrail), GCP (Compute Engine, Cloud Run, VPC), Docker, Terraform, GitHub Actions, OPA/Rego/Conftest, LocalStack
OS: Linux, Windows